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The occurrence and enrichment mechanism of indium in sphalerite from Dachang Sn polymetallic ore field, South China: Part I. The Tongkeng deposit

Shi-Ni Fan ^{a, b}, Yu-Miao Meng ^{a, *}, De-Feng Shi ^b, Xiao-Wen Huang ^a, Qi-Nan Zou ^{a, c}, Meng-Jie Jia ^{a, b}

^a State Key Laboratory of Critical Mineral Research and Exploration, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang 550081, China

^b School of Earth Sciences and Spatial Information Engineering, Hunan University of Science and Technology, Xiangtan 411201, China

^c University of Chinese Academy of Sciences, Beijing 100049, China

* Corresponding author

E-mail address: mengyumiao@vip.gyig.ac.cn (Yu-Miao Meng)

Abstract: Indium, a dispersed and strategically important element, typically exists as an associated element within various polymetallic deposits. The Dachang ore field, renowned for its substantial Sn reserves, also hosts significant high-grade In resources. However, the occurrences and enrichment mechanisms of In in this field remain unclear.

The Tongkeng Sn polymetallic deposit of the Dachang ore field in Guangxi was chosen as an example to decipher the above issues. Three generations of sphalerite (Sp1-Sp3) were identified based on different mineral assemblages. Sp1 is mainly associated with chalcopyrite and galena, whereas Sp2 is always associated with cassiterite and stannite. Sp3 mainly coexists with cassiterite and pyrite. The average In contents of Sp1, Sp2, and Sp3 are about 99, 954 and 689 ppm, respectively. Sp1 is enriched in Fe, Cd, Pb, and Bi, while Sp2 and Sp3 are predominantly enriched in In, Cu, Ga, and Sn. Indium may be present in sphalerite via the substitution of $\text{In}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ and $(\text{In}^{3+}, \text{Sn}^{3+}) + (\text{Cu}^+, \text{Ag}^+) \leftrightarrow 2\text{Zn}^{2+}$. Copper exhibits a consistent positive correlation with In enrichment across all sphalerite generations, while the effects of Ag and Sn show generation-specific variations. A significant increase in In content of sphalerite when Cd content ranges from 0.35 wt% to 0.45 wt%, consistent with the "Indium Window Effect". Using the GGIMFis geothermometer, the formation temperatures for Sp1, Sp2, and Sp3 were estimated to be 329-392°C, 321-354°C, and 325-357°C, respectively. The relationship between In content and temperature indicates that the effect of temperature on In enrichment is content-dependent and should be evaluated within specific compositional ranges.

Keywords: Indium; Occurrence; Enrichment mechanism; Tongkeng deposit; Dachang

1. Introduction

Indium (In) is a silvery-white metal known for its excellent conductivity, light permeability, low resistance, high plasticity, and good ductility (Wang, 2015). These properties make it widely utilized in high-tech and electronic industries. Indium exhibits a highly heterogeneous global distribution, with its resources predominantly concentrated in Sn-bearing ore belts along continental margins, including Central Europe, southern Australia, South America, East Asia, and eastern Canada (Zhang et al., 2019). Due to its scarcity, In has been classified as a critical strategic mineral resource in Europe and North America, prompting targeted resource security strategies (Zhang et al., 2019). As the world's largest producer of In, China faces an increasing supply-demand imbalance due to the rising global demand for In resources (Li et al., 2020). Understanding the enrichment mechanism of In is thus crucial for ensuring the sustainable development of In resources worldwide.

Indium is a dispersed element, which is difficult to form independent minerals and deposits, typically exists in association with other metal deposits (Zhang et al., 2003; Zhang et al., 2008; Xu and Li, 2018). Numerous studies have examined the types of deposits containing In resources, as well as the occurrences and enrichment mechanisms of In (Ishihara et al., 2006; Werner et al., 2017; Mejías et al., 2023; Zheng et al., 2023). Indium can be present in the form of independent minerals, nanoparticles, and isomorphic substitution. A total of 19 independent minerals of In have been discovered, among which roquesite is the most discovered independent mineral of In (Xu and Li, 2018; Zheng et al., 2023). Sphalerite is one of important hosts for In (Bauer et al., 2019). McIntyre et al. (1984) found that Ag and In in sulfides always replace Zn in pairs, which is the first evidence to obtain the substitution mechanism of In in sphalerite. Indium, together with other trace elements, also can replace Zn in sphalerite via the ways of $\text{In}^{3+} + \text{Cu}^+ \rightarrow 2\text{Zn}^{2+}$ and $\text{In}^{3+} + \text{Sn}^{2+} + \text{Ag}^+ \rightarrow 3\text{Zn}^{2+}$ (Cook et al., 2009; Ciobanu et al., 2011; Frenzel et al., 2016; Xu et al., 2021). In addition, the substitution mechanism of In would be influenced by fluid sources, properties, and physicochemical conditions. For example, the degree of mantle melting controls the content of In in the magma source (Xu and Li, 2018), and temperature can affect the enrichment of In in sphalerite (Li et al., 1984). By summarizing the substitution forms of In, it is believed that the overall formula is as follows: $\text{M}^+ + \text{M}^{3+} \rightarrow 2\text{Zn}^{2+}$ and $(x+2y) \text{M}^+ + y\text{M}^{2+} + x\text{M}^{3+} + y\text{M}^{4+} \rightarrow (4-4y-2x) \text{Zn}^{2+}$ ($\text{M}^+ = \text{Ag}, \text{Cu}$; $\text{M}^{2+} = \text{Cu}, \text{Fe}, \text{Cd}, \text{Hg}, \text{Co}, \text{Ge}$; $\text{M}^{3+} = \text{In}, \text{Ga}, \text{Fe}, \text{Tl}, \text{Sb}, \text{Sn}$; $\text{M}^{4+} = \text{Ge}, \text{Sn}, \text{Mo}, \text{W}$) (Johan, 1988; Belissont et al., 2014; Xu and Li, 2018).

The primary In deposits in China include skarn-type and polymetallic vein-type Sn-Zn polymetallic deposits mainly associated with magmatic hydrothermal processes (Zhang et al., 1998; Zhang et al., 2003). Currently, the identified In resources are predominantly located in the South China fold belt, which are the most significant Sn-In polymetallic deposits, such as Dachang in Guangxi, Dulong and Gejiu in Yunnan (Li et al., 2020). The Shizhuyuan and Xianghualing ore fields in southern of Hunan province also contain notable grades of In (Liu et al., 2017a; Liu et al., 2017b). The

Dachang Sn polymetallic ore field is globally recognized as a large to ultra-large Sn deposit, with In resource reserves of up to 8,775 tons. Previous studies are mainly focused on ore genesis, mineralization ages, sources of ore-forming materials of deposits, and so on in this ore field (Zhao et al., 2018; Hu et al., 2020b; Li et al., 2021; Cai et al., 2023; Li and Zhou, 2023). However, researches on the occurrences and enrichment mechanisms of In in sphalerite are relatively limited (Luo et al., 2009; Dai et al., 2012; Pi et al., 2015). Luo et al. (2009) measured In contents across different deposits in the Dachang ore field and observed a gradual decrease in In content radiating outward from the Longxianggai granite body, with enrichment increasing at greater depths. Dai et al. (2012) revealed elevated In concentrations (average 1.88 ppm) within the ore-bearing strata and a positive correlation between In content and temperature along the vertical profile. Pi et al. (2015) documented a depth-dependent decrease in In content in the Tongkeng deposit, whereas the Gaofeng deposit exhibited relatively stable In concentrations with depth.

In this study, the Tongkeng deposit in the western part of the Dachang ore field was studied as an example to investigate the occurrence and enrichment mechanism of In in Sn polymetallic deposits. Detailed petrography combined with in situ element analysis techniques such as electron probe microanalysis (EPMA) and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) were used to characterize the distributions and variations of In in sphalerite from different mineralization generations. The factors controlling enrichment of In in sphalerite were also discussed in detail.

2. Regional geology

The Youjiang Basin is situated at the convergence of western Guangxi, southeastern Yunnan, and southwestern Guizhou, tectonically aligned along the southern margin of the Yangtze Plate (Feng et al., 2023). This basin contains several significant Sn polymetallic deposits, predominantly located in the northwestern region, including Dulong, Gejiu, and Bainiuchang. Additionally, numerous deposits were also found in the northeastern part, such as Mangchang, Dachang, Wuxu, and Damingshan (Fig. 1a).

The Dachang Sn polymetallic ore field in Guangxi is situated on the southwestern edge of the Jiangnan ancient land, on the northeastern side of the Youjiang Basin, and centrally located within the Ziyun-Du'an Fault Zone (Fig. 1a). The exposed strata primarily consist of carbonate sedimentary rocks from the Devonian, Carboniferous, Permian and Triassic periods, as well as Quaternary sediments (Shao, 2019; Li and Zhou, 2023). The lithology of the Devonian period mainly includes sandstone, shale, limestone, and siliceous rock. The lower section of the Carboniferous is predominantly composed of clastic rocks, while the middle and upper sections are characterized by limestone and dolomitic limestone. The Permian strata are primarily made up of

siltstone, sandstone, calcareous mudstone, limestone, and dolostone. The Triassic strata are largely composed of mudstone, limestone, shale, and sandstone. The Quaternary comprises residual, slope, and alluvial deposits (Tang, 2017; Xie, 2017; Zheng, 2018; Zhou, 2019). Within the mining area, there are fold structures trending northwest (NW) and nearly north-south (NS), including the Dachang anticline, Bamianshan anticline, and Longxianggai anticline. Additionally, fault structures trend NW, northeast (NE), and nearly south-north (SN), with the NW-trending faults primarily represented by the Dachang Fault and Danchi Fault, while the NE-trending faults include the Tongkeng and Heishuigou faults (Hu et al., 2020a; Li and Zhou, 2023).

The central Longxianggai complex is the primary magmatic rocks, which is the largest granite body in this region. This complex appears as rock beds, walls, and branches, which extend deep into the Tongkeng deposit (Liang et al., 2007; He et al., 2024). On the western side, the magmatic rocks are mainly located along both sides of the Tongkeng deposit, forming a substantial near north-south (NS) trending rock wall, referred to as the "Eastern Rock Wall" (Cretaceous granite porphyry) and the "Western Rock Wall" (Cretaceous diorite porphyry) (Guo et al., 2020).

The Tongkeng deposit is distributed in a ring-shaped pattern around the central Longxianggai granite mass. Based on their spatial distribution, the mineralized areas are categorized into three mineral belts: west, central, and east (Fig. 1b). The western belt includes the Tongkeng-Changpo, Gaofeng, and Bali-Longtoushan Sn polymetallic deposits. The central belt comprises the Lame Zn-Cu deposit and the Chashan W-Sb deposit. The eastern belt features the Dafulou, Kangma, and Huile Sn polymetallic deposits.

3. Deposit geology

The Tongkeng deposit is situated in the western ore belt of the Dachang Sn polymetallic ore field in Guangxi, and it represents the largest Sn polymetallic deposit within this ore field. It primarily consists of the No. 91 and No. 92 Sn polymetallic orebodies, as well as the No. 94, 95, and 96 skarn-type Zn-Cu orebodies (Fig. 2). The exposed strata include the Middle Devonian Nabiao Formation characterized by mudstone-limestone and reef limestone, the Luofu Formation consisting of siltstone, mudstone, and mud-limestone, the Upper Devonian Liujiang Formation which features siliceous rock, the Wuzhishan Formation of limestone, and the Tongchejiang Formation comprising mudstone, along with Carboniferous limestone. The deposit is primarily influenced by NW-trending structures such as the Dachang anticline and Dachang fault, along with the NE-trending Tongkeng fault. Notably, both the NW-trending Dachang anticline and the Dachang fault exhibit a near east-west (EW) orientation (Xu et al., 2012; Pi et al., 2015; Hu et al., 2022). The ore bodies are controlled by stratigraphic levels and faults, occurring mainly in layered, bedded, vein-like, and stringer forms (Xu et al., 2012; Cai et al., 2020; Cai et al., 2023).

The Tongkeng deposit consists of upper Sn polymetallic ore bodies and lower Zn-Cu ore bodies. The upper Sn polymetallic ore bodies primarily contains ore minerals such as cassiterite, sphalerite, pyrite, marcasite, arsenopyrite, pyrrhotite, and small amounts of jamesonite, galena, chalcopyrite, stannite, tetrahedrite, magnetite, ferberite, scheelite, tungstenite, galenobismutite, bismuthinite, and laitakarite. The gangue minerals include quartz, calcite, tourmaline, kutnohorite, sericite, fluorapatite, titanite, rutile, and fluorite (Fig. 3). In the lower Zn-Cu ore bodies, the primary ore minerals are sphalerite, chalcopyrite, pyrite, arsenopyrite, and pyrrhotite, while the gangue minerals consist of quartz, calcite, tourmaline, and a minor amount of skarn minerals (Fig. 3) (Cai et al., 2005; Zhang, 2019; Cai et al., 2023). Widespread alteration of the surrounding rock is evident, with alterations related to the lower Zn-Cu ore body mainly encompassing skarnization, silicification, and tourmalinization. In contrast, alterations associated with the upper Sn polymetallic ore body predominantly include silicification, tourmalinization, sericitization, and carbonatization (Zhang et al., 2019; Li and Zhou, 2023).

The mineralization process of the Tongkeng deposit can be divided into two periods and five stages (Zhang, 2019; Hu, 2022; Cai, 2023). Based on the previous studies and our petrographic observation, the mineralization periods are divided into hydrothermal Zn-Cu mineralization period includes the skarn and the sulfide stages. Cassiterite-sulfide mineralization period includes cassiterite-arsenopyrite-pyrrhotite-pyrite-quartz, cassiterite-sphalerite-jamesonite-quartz, and jamesonite-quartz-calcite stages. Among these, cassiterite-sulfide mineralization period is identified as the primary mineralization period (Fig. 3).

4. Samples and analytical methods

4.1. Sampling

Based on the observations of under the mine, it was determined that the Tongkeng deposit is irregularly filled with limestone and siliceous rock through fissures (Fig. 4a), and also exhibits a layered formation within the siliceous rock (Fig. 4b). The surrounding rock primarily consists of siliceous rock and limestone. The samples analyzed in this study were collected from of the No. 91 body of Tongkeng deposit, mainly through the mining engineering tunnels, with sampling conducted sequentially from top to bottom according to the elevation from 355m to 305m level of the ore body. Detailed sample information was provided in Appendix Table S1.

4.2. EPMA

Major elements and some trace elements analysis of sphalerite was conducted at the State Key Laboratory for Critical Mineral Research and Exploration, Institute of

Geochemistry, Chinese Academy of Sciences. A JXA8530F-plus EPMA was used with an acceleration voltage of 25.0 kV, a beam diameter of 10 μm , and a current of 10 nA. Standards included pyrite (Fe), sphalerite (Zn), chalcopyrite (Cu), metallic silver (Ag), metallic selenium (Se), sphalerite (S), indium phosphide (In), metallic cadmium (Cd), and cassiterite (Sn).

4.3. LA-ICP-MS

Trace element analysis of sphalerite was also performed at the State Key Laboratory for Critical Mineral Research and Exploration, Chinese Academy of Sciences, using LA-ICP-MS. The laser ablation system is the ASI RESOLUTION-LR-S155 excimer laser, while ICP-MS is the Agilent 7700x. Argon gas (at 900 ml/min) and helium gas (at 350 ml/min) were mixed in the ablation chamber before entering the ICP along with the sample aerosol.

Typical working parameters for the laser included a frequency of 5 Hz, an energy output of 2.5-3 J/cm², and a spot size of 26 μm . Prior to analysis, SRM610 was used to optimize the performance of the ICP-MS, achieving optimal sensitivity and ionization efficiency ($\text{U/Th} \approx 1$), the lowest oxide yield ($\text{ThO/Th} < 0.3\%$), and a minimal background value. Each acquisition cycle consisted of approximately 30 seconds of blank signal followed by 60 seconds of sample signal. Concentration calculations employed multiple external standards and total normalization methods (Liu et al., 2008). The external standards included STDGL3, GSD-1G, and naturally pure pyrite, along with the sulfide international standard MASS-1, which served as quality control samples to monitor data reliability. Among these, STDGL3 was used to correct for chalcophile and siderophile elements, while GSD-1G was employed for lithophile elements. The analysis precision of the elements with contents higher than 10 ppm is about 5-10% (Danyushevsky et al., 2011). The analyzed isotopes included ²⁹Si, ⁴³Ca, ⁴⁵Sc, ⁴⁹Ti, ⁵¹V, ⁵³Cr, ⁵⁵Mn, ⁵⁷Fe, ⁵⁹Co, ⁶⁰Ni, ⁶⁵Cu, ⁶⁶Zn, ⁷¹Ga, ⁷²Ge, ⁷⁴Ge, ⁷⁵As, ⁷⁷Se, ⁸²Se, ¹⁰⁷Ag, ¹⁰⁹Ag, ¹¹¹Cd, ¹¹³Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ¹²⁵Te, ¹⁹⁷Au, ²⁰⁵Tl, ²⁰⁸Pb and ²⁰⁹Bi.

5. Results

5.1. Petrography

Based on the mineral assemblages, sphalerite can be divided into three generations (Sp1, Sp2, and Sp3), which correspond to sulfide stage in the hydrothermal Zn-Cu mineralization period, cassiterite-quartz stage, and cassiterite-sulfide-quartz stage in the cassiterite-sulfide mineralization period, respectively (Fig. 3).

Sp1 was from the massive and nodular ores represented by samples TK24-55 and TK24-16 (Fig. 4c, d). These ores mainly comprise sphalerite, pyrrhotite, chalcopyrite,

and minor galena (Fig. 5a, b). Chalcopyrite inclusions and veins were commonly contained in Sp1 (Fig. 5a, b). BSE images revealed that sphalerite from the Sp1 stage primarily coexists with chalcopyrite, pyrrhotite, and pyrite, exhibiting significant internal dissolution texture and structure (Fig. 6a, b).

Sp2 was from the massive ores represented by samples TK24-24 (Fig. 4e). The primary mineral composition mainly consists of cassiterite and sphalerite, followed by stannite. Inside the sphalerite, smaller mineral particles such as chalcopyrite and pyrrhotite also can be observed (Fig. 5c, d), while the vein minerals predominantly comprise siderite and quartz. Sphalerite is associated with extensive amounts of cassiterite, where numerous stannite are present at the contact points between sphalerite and cassiterite; additionally, large areas of stannite can also be found within the sphalerite itself (Fig. 6c, d).

Sp3 was from the massive ores represented by sample TK24-4 (Fig. 4f). The primary mineral composition includes sphalerite, followed by pyrite, pyrrhotite, and cassiterite (Fig. 5e, f), while the gangue minerals are predominantly quartz and calcite. Sphalerite from the Sp3 stage coexists with cassiterite and pyrite, indicating that this mineralization phase occurs slightly later than Sp2 (Fig. 6e, f). In addition, there are minerals such as ferberite, wolframite, and bismuthinite, further highlighting the diverse range of mineral species present in the Tongkeng deposit.

5.2. Major element composition of sphalerite

The EPMA results are summarized in Table 1, and full results are listed as Appendix Table S2. Sphalerite contains a certain amount of trace amounts including Cd, In, and Cu. The contents of Se, Ag, and Sn are generally below the detection limits.

The average contents of Zn, Fe, and S in Sp1 is 53.06%, 12.05%, and 33.91%, respectively. The content of In is below the detection limit, and no variations in content are observed. The average content of Zn in Sp2 is 53.54%, the average content of Fe is 11.64%, the average content of S is 33.71%, and the average content of In is 0.14%. The average contents of Zn, Fe, and S is 52.83%, 12.61%, and 33.78%. The average content of In is 0.13%. Comparisons show that the content variations of Fe, Zn, and S in the three generations of sphalerite are similar, and the contents of In in Sp2 and Sp3 are similar, but which are significantly higher than Sp1.

Based on the EPMA results, two highest In content areas in Sp2 are chosen for further element mapping analyses. Elements such as Cu, In, and Cd within sphalerite are uniformly distributed (Fig. 7a, b). The X-ray mapping of EPMA showing the coexistence of sphalerite and cassiterite (Fig. 7a) illustrates that Zn, S, and Fe are evenly distributed throughout sphalerite, while Sn is uniformly distributed in cassiterite. Notably, the Cu content is significantly higher at the edges of sphalerite than within its core, creating a ring-like appearance. Furthermore, In and Cd are present in low

concentrations and are evenly distributed within sphalerite. In the EPMA mapping depicting the coexistence of sphalerite and stannite (Fig. 7b), Zn, S, and Fe maintain uniform distribution in sphalerite, while Sn and Cu are uniformly distributed in stannite. Both In and Cd exhibit low concentrations with uniform distribution in both sphalerite and stannite. Additionally, it is observed that the In content in stannite is slightly higher than that found in sphalerite.

5.3. Trace element composition of sphalerite

The maximum, minimum, and average contents are summarized in Table 2 and full results are listed as Appendix Table S3. Additionally, the elemental contents of the three generations of sphalerite are illustrated in box plots (Fig. 8).

Sp1 is relatively rich in Co, Cu, Ga, and Sn, whereas Sp2 has higher Cu, Ga, In, and Sn contents. Sp3 is characterized by relatively high Bi content. Sp1 has lower In content of 19-308 ppm averaging at 99 ppm. Sp2 has In content ranging from 677 to 1433 ppm (average content = 954 ppm). The In content of Sp3 varies from 480 to 1127 ppm (average content = 689 ppm). Overall, the trend of the average In contents of the three generations of sphalerite shows the trend of $Sp2 > Sp3 > Sp1$.

LA-ICP-MS mappings of sphalerite grain (Sp2) show that In and Ga are heterogeneously distributed and Fe, Cu, Cd, Sn, Ag, Sb, Pb, and Bi are relatively homogenous (Fig. 9). Another sphalerite grain (Sp2) shows that the major elements in sphalerite, such as Zn, S, and Fe, are uniformly distributed. In is relatively evenly distributed in both sphalerite and stannite; however, its concentration is significantly higher in stannite than in sphalerite. In contrast, Cd exhibits an opposite distribution pattern, displaying a notably higher concentration in sphalerite compared to stannite. Cu and Sn are found in greater amounts within stannite and show a uniform distribution, while silver Ag demonstrates an irregular distribution in stannite (Fig. 10).

In order to better understand the compositional differences between different generations of sphalerite, we utilized Origin software to conduct the principal component analysis (PCA), applying the relevant matrix to extract the principal components. Sp1 can be separated from Sp2 and Sp3 due to its positive correlation with Co, Mn and Fe (Fig. 11a, b). However, Sp2 and Sp3 cannot be clearly separated in the plot of PC1-PC2 because both of them show positive correlation with Ag, In, Cu, Ga, and Sn (Fig. 11a, b).

6. Discussion

6.1. The occurrences of trace elements in sphalerite

Using LA-ICP-MS to analyze mineral composition allows us not only to determine the concentrations of elements but also to understand their spatial variations with different depths of ablation, providing us a foundation for discussing the occurrence of these elements (Fan et al., 2018). It is generally accepted that a relatively flat laser ablation profile for an element indicates its primary existence in the form of an isomorphism within the mineral lattice. In contrast, a laser ablation profile exhibiting irregular or multiple peaks normally suggests the presence of the element as microscopic inclusions (Cook et al., 2009; Ye et al., 2011). Numerous studies have demonstrated that sphalerite serves as a host mineral for various trace elements, with divalent metal cations such as Fe^{2+} , Mn^{2+} , Cd^{2+} , Co^{2+} , and Sn^{2+} readily substituting for Zn^{2+} in the sphalerite structure. There are still monovalent and trivalent substitutes for divalent (Murakami and Ishihara, 2013; Belissont et al., 2014; Lockington et al., 2014).

In the time-resolved profiles obtained from LA-ICP-MS, the signals for Mn, Fe, In, and Cd are relatively stable (Fig. 12a-d), indicating that these elements primarily exist in sphalerite as isomorphic substitution. In contrast, elements such as Cu, Sn, Ag, Sb, Pb, and Bi exhibit obvious peaks in the time-resolved profile (Fig. 12b-d), suggesting that these elements are predominantly present as microscopic inclusions. The co-variation in the peaks of Cu, Sn, and Ag (Fig. 12b) indicates that these elements likely exist as inclusions of chalcopyrite, while the co-variation of Pb, Bi, and Sb peaks (Fig. 12d) suggests their presence as inclusions of galena. This observation is consistent with numerous chalcopyrite and galena inclusions within the sphalerite under the observation of petrography (Fig. 5b, c). It is worth mentioning that the signal line of In element on the time-resolved profiles is relatively stable (Fig. 12a-d), but under LA-ICP-MS mapping (Fig. 9), it can be seen that the distribution of In in sphalerite is uneven, indicating that even if In exists in the form of isomorphic in sphalerite, it still exhibits the characteristic of uneven distribution.

Substitution ways of elements in sphalerite can be inferred from correlation between elements. A strong positive correlation between In and Cu across all three generations of sphalerite (Fig. 13a), suggesting a potential substitution mechanism of $\text{In}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ (Cook et al., 2009). Furthermore, when the Cu content in sphalerite exceeds 1000 ppm, the In content deviates from the general trend line, indicating that these high-Cu points likely reflect chalcopyrite inclusions within sphalerite. This is consistent with amounts of exsolved chalcopyrite in sphalerite (Fig. 5c). The positive correlation between Cu + Ag and In in both Sp1 and Sp3 (Fig. 13b), indicating that In can be incorporated into sphalerite via $\text{In}^{3+} + (\text{Cu}^+, \text{Ag}^+) \leftrightarrow 2\text{Zn}^{2+}$. There is a positive correlation between Cu+Ag and Sn+In in Sp1, Sp2, and Sp3 (Fig. 13c), indicating another possible substitution mechanism of $(\text{In}^{3+}, \text{Sn}^{3+}) + (\text{Cu}^+, \text{Ag}^+) \rightarrow 2\text{Zn}^{2+}$. In addition, it also can be seen that the In and Sn show a weak positive correlation among the three generations of sphalerite, and the effects of the three generations of sphalerite are different (Fig. 13d).

6.2. The effect of trace elements on In enrichment in sphalerite

The correlation coefficients (R^2) of Sp1, Sp2, and Sp3 are 0.75, 0.69 and 0.85, respectively, indicating that the In is coupled with Cu for all generations of sphalerite. It is believed that the presence of Cu promotes the enrichment of In in sphalerite (Fig. 13a). Similarly, fitting analysis was also conducted on other elements that showed positive correlations with In. Except Cu, the presence of Ag also slightly affects the enrichment of In in Sp2, but had little effect on the enrichment of In in Sp1 and Sp3, and even slightly inhibited it (Fig. 13b). The presence of Sn is similar to that of Ag, with a promoting effect on the enrichment of In in Sp2. The difference is that it has a more significant promoting effect on the enrichment of In in Sp1, but has a negative effect on Sp3 (Fig. 13c).

Previous studies have proposed that In and Sn migrate together in ore-forming hydrothermal fluids and separate during subsequent mineralization processes when the sulfide appears in large quantities, suggesting that Sn plays a decisive role in the enrichment of In in the initial stage (Zhang et al., 2003). The fitting analysis of Sn and In in three generations of sphalerite indicates that the relationship between the two elements in sphalerite is not significant (Fig. 13d). But this relationship should be considered with caution because Sn content is one order of magnitude lower than In. The content of Mn in sphalerite is relatively high and varies greatly. The correlation between Mn and In in Sp1 is not significant, while in Sp2 and Sp3, Mn and In show a negative correlation. It is speculated that the increase in Mn content will inhibit the enrichment of In in sphalerite (Fig. 13e).

The In content in sphalerite from the Tongkeng deposit shows a significant increase when Cd content falls between 0.35 wt% and 0.45 wt% (Fig. 13f), which is similar to the observation of Pi et al. (2015). Extreme In enrichment at a specific range of Cd content has been first reported the Argentina shallow hydrothermal Au-Cu-Zn-Pb-Ag deposits. In this deposit, when the Cd content in sphalerite ranged from 0.2 wt% to 0.6 wt%, the In content significantly increased, which was defined as "Indium Window Effect" (Dill et al., 2013). Liu et al. (2017b) suggested that a similar effect exists at the Xianghualing Sn-Zn deposit, Hunan, where In content notably increases when Cd content in sphalerite is between 0.35 wt% and 0.45 wt%. The origin of the "Indium Window Effect" has been attributed to structural transformations during ZnS precipitation. Dill et al. (2013) proposed that under specific physicochemical conditions, the cubic crystal system of sphalerite (ZnS) may undergo a phase transition into the tetragonal structure of roquesite (CuInS_2). This transformation is thought to be a key factor in the development of the "Indium Window Effect," as it facilitates the selective enrichment and incorporation of indium into the mineral lattice. Duan et al. (2024) calculated the $^{115}\text{In}/^{113}\text{In}$ ratio in indium containing minerals using first-principles calculations and found that the chemical composition of sphalerite may affect the indium isotope fractionation factor by altering the bond length of indium. The specific influencing factors and control mechanisms of the "Indium Window Effect" in this study are still unclear and require further research.

6.3. The effect of temperature on In enrichment

Previous studies have demonstrated that temperature can play a significant role in the enrichment of trace elements in sphalerite. As a result, the concentrations of specific elements and their associated ratios (such as Zn/Fe, Zn/Cd, Cd/Fe, and Ga/In) in sphalerite can serve as indicators of the metallogenic temperature (Liu et al., 1984; Han, 1994; Cook et al., 2009; Zou et al., 2012; Belissont et al., 2014; Bauer et al., 2018; Kang et al., 2020; Zhang et al., 2021). Sphalerite formed in high-temperature environments (200-350 °C) tends to be darker in color and shows higher concentrations of Fe, Mn, Co, and In elements, while sphalerite formed under low-temperature conditions (110-180 °C) is characterized by higher concentrations of Cd, Ga, and Ge elements (Cao et al., 2014; Wang et al., 2021; Chen et al., 2023; Meng et al., 2024). Specifically, the sphalerite formed under high temperature conditions exhibits content ratios of Zn/Fe<10, Zn/Cd>500, Cd/Fe<0.02, and Ga/In<0.01. In contrast, sphalerite formed under low-temperature conditions displays: Zn/Fe > 100, Zn/Cd < 100, Cd/Fe > 1, and Ga/In ranging from 1 to 100. Sphalerite formed at intermediate temperatures falls between these two end members (Table 3).

To better estimate the crystallization temperature of the three generations of sphalerite, the formula proposed by Frenzel et al. (2016) was applied. The calculation results are presented in Table 3, where the formation temperature of Sp1 ranges from 329°C to 392°C (average 363°C), the formation temperature of Sp2 ranges from 321°C to 354°C (average 338°C), and the formation temperature of Sp3 ranges from 325°C to 357°C (average 343°C). These results are consistent with results of fluid inclusion thermometry (260–360°C) (Li et al., 2021), which suggested that all three generations of sphalerite formed in an intermediate to high-temperature hydrothermal environment.

Previous studies have indicated a positive correlation between the In content in sphalerite and the formation temperature within the range of 280 to 350°C (Dai et al., 2012). When In content in Sp1 is less than 400 ppm, a positive correlation between In content and temperature indicates that the enrichment of In is significantly controlled by temperature (Fig. 14). However, when In contents in Sp2 and Sp3 exceed 400 ppm, a negative correlation between In content and temperature was observed (Fig. 14). Therefore, the In enrichment was not solely controlled by temperature, and the relationship between In enrichment and temperature should be discussed at a specific concentration range.

7. Conclusions

Petrography and trace element geochemistry of sphalerite from the Tongkeng deposit were investigated to reveal the occurrence and enrichment mechanism of In. The average content of In in the Sp1, Sp2, and Sp3 of the Tongkeng deposit is 99, 954,

and 689 ppm, respectively. Indium is mainly present in sphalerite in the form of isomorphism, and maybe influenced by the coupling substitution mechanism via $\text{In}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ and $(\text{In}^{3+}, \text{Sn}^{3+}) + (\text{Cu}^+, \text{Ag}^+) \leftrightarrow 2\text{Zn}^{2+}$. Copper plays a significant role in promoting In enrichment across all sphalerite generations. The effects of Ag and Sn on In enrichment vary by generation. Additionally, the "Indium Window Effect" is observed in the Tongkeng deposit. The influence of temperature on In enrichment is concentration-dependent, and can be positive or negative within a specific compositional range.

CRediT authorship contribution statement

Shi-Ni Fan: Conceptualization, Investigation, Visualization, Writing - original draft. **Yu-Miao Meng:** Investigation, Funding acquisition, Project administration, Writing - review & editing. **De-Feng Shi:** Investigation, Visualization, Writing - review & editing, Supervision. **Xiao-Wen Huang:** Methodology, Formal analysis, Writing - review & editing. **Qi-Nan Zou:** Investigation, Visualization. **Meng-Jie Jia:** Investigation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figure captions

Fig. 1. (a) Distribution map of Sn polymetallic deposits in the Youjiang Basin, South China (modified from [Guo et al., 2018](#); [Zhao et al., 2018](#)); (b) Geological map of the Dachang Sn polymetallic ore field (modified from [Hu et al., 2020a](#)).

Fig. 2. Schematic diagram of geological profile the Tongkeng deposit (modified from [Hu et al., 2020b](#)).

Fig. 3. Paragenetic sequence of minerals in the Tongkeng deposit (modified from [Cai, 2023](#)).

Fig. 4. Field photos and photomicrographs of some representative ores from the Tongkeng deposit. (a) Vein-type sulfide ore body in limestone; (b) Lens-shaped Zn-Cu ore body and veinlets in siliceous rock; (c) Massive Zn-Cu ore consisting of sphalerite, pyrrhotite, and minor galena (TK24-55); (d) Massive Zn-Cu ore composed of pyrrhotite, sphalerite, and chalcopyrite (TK24-16); (e) Massive tin ore mainly composed of cassiterite, pyrite and sphalerite (TK24-24); (f) Banded ore consisting of pyrite, sphalerite, and pyrrhotite (TK24-4). Abbreviations: Gn, Galena; Sp, sphalerite; Po, Pyrrhotite; Ccp, chalcopyrite; Py, Pyrite; Cst, Cassiterite.

Fig. 5. Photomicrographs of representative ores from the Tongkeng deposit. (a) Inclusion-rich sphalerite (Sp1) associated with chalcopyrite; (b) Chalcopyrite veins cuts through massive sphalerite (Sp1) associated with galena and pyrrhotite; (c-d) Inclusion-rich sphalerite (Sp2) associated with cassiterite and stannite. Stannite shows replacement relics structure; (e) Sphalerite (Sp3) associated with anhedral pyrite and subhedral pyrrhotite; (f) Sphalerite showing a textural equilibrium with coarse-grained cassiterite. Abbreviations: Sp, sphalerite; Ccp, chalcopyrite; Po, Pyrrhotite; Stn, Stannite; Gn, Galena; Cst, Cassiterite; Py, Pyrite.

Fig. 6. Back-scattered electron (BSE) images showing mineral assemblages in the Tongkeng deposit. (a) Pyrite and chalcopyrite inclusions in sphalerite (Sp1); (b) Sphalerite (Sp1) associated with chalcopyrite, pyrite, and pyrrhotite; (c) Sphalerite (Sp2) and stannite crosscut by chalcopyrite and pyrite veins; (d) Sphalerite (Sp2) associated with cassiterite and quartz; (e) Spaces between sphalerite (Sp3) and pyrrhotite infilled by pyrite and fluorite; (f) Sphalerite (Sp3) associated with quartz and cassiterite. Abbreviations: Sp, sphalerite; Ccp, chalcopyrite; Py, Pyrite; Stn, Stannite; Cst,

Cassiterite; Po, Pyrrhotite; Qtz, Quartz; Sd, Siderite; Fl, Fluorite.

Fig. 7. EPMA element distribution maps of sphalerite (Sp₂). Abbreviations: Sp, sphalerite; Cst, Cassiterite; Stn, Stannite; Sd, Siderite.

Fig. 8. Box plots for selected elements in sphalerite from the Tongkeng deposit. The box for each element represents the 25% (lower margin) to 75% (upper margin) data ranges. The square and horizontal line represent the mean and median, respectively.

Fig. 9. LA-ICP-MS element maps showing the distribution of selected elements in the sphalerite and cassiterite of the Tongkeng deposit.

Fig. 10. LA-ICP-MS element maps showing the distribution of selected elements in the sphalerite and stannite of the Tongkeng deposit.

Fig. 11. Principal component analysis of the log-transformed LA-ICP-MS data of sphalerite from the Tongkeng deposit. (a) Score plot for different generations of sphalerite; (b) Loading plot for elements. Data in parentheses represent the proportion of each principal component explaining the variance.

Fig. 12. Representative time-resolved LA-ICP-MS depth profiles for sphalerite from the Tongkeng deposit. (a) Relatively smooth signals for analyzed elements in sphalerite without contamination of inclusions; (b-c) Stannite inclusion in sphalerite; (d) Galena and stannite inclusion in sphalerite.

Fig. 13. Binary plots of Cu vs. In (a), Cu+Ag vs. In (b), Cu+Ag vs. Sn+In (c), Ga vs. In (d), Sn vs. In (e), Cd vs. In (f) in sphalerite from the Tongkeng deposit.

Fig. 14. Plot of In contents versus temperature of three generations of sphalerite. The temperature density distribution of three generations of sphalerite is shown in the upper part of the plot, and the In content density distribution of them are shown on the right side of the plot.

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